Sabbir Ahmed

sabbir.ahmed.cse19.just@gmail.com sabbbir.github.io
GitHub Kaggle Hugging Face

Education

M.Sc in Computer Science and Engineering *Expected: Aug 2027*
Jashore University of Science and Technology *Jashore, Bangladesh*
CGPA: 3.69/4.00 (*Upto 1st Semester*)
Thesis: Robust Fine-Tuning and Alignment of Large Language Models

B.Sc in Computer Science and Engineering *2025*
Jashore University of Science and Technology *Jashore, Bangladesh*
CGPA: 3.20/4.00
Thesis: Guided Unsupervised Segmentation via Cross-Model Prompting

Research Interest

Computer Vision, Large Vision-Language Models, Multi modal Learning

Research Experience

Research Assistant *Jul 2024 – Present*
Bangladesh Computational Social Science Lab *Remote*
PI: Dr. Nur Ahmed, Digital Fellow at MIT, Assistant Professor at University of Arkansas

- Developed an **LLM-based** framework to study institutional DEI discourse across **500 U.S. universities**, classifying **700,000 tweets** via parallel Gemini APIs within a 9-million-tweet corpus.
- Fine-tuned and evaluated BERTweet and **Qwen models (4B, 9B, 14B, 27B)** using class-balanced samples, including **Qwen3.6-27B** trained on **4,500 text** examples with generated **reasoning traces.**
- Engineered distributed **Qwen3.6-27B** inference across **cloud servers**, integrating Hugging Face for adaptive batching and fault-tolerant worker coordination at scale.

Research Assistant *2023 – Present*
Visual Machine Intelligence Lab, JUST *Jashore, Bangladesh*
PI: A F M Shahab Uddin, PhD, Assistant Professor, JUST

- Conducted empirical benchmarking of **image mixing augmentations** across **273 experiments** on CIFAR/ImageNet, evaluating augmentation strategies across diverse architectures.
- Designed **CutMix, SaliencyMix, MixUp, and PuzzleMix** variants through controlled experiments, investigating intra-class and inter-class mixing mechanisms affecting model performance.
- Analyzed performance trends, feature representations, and generalization patterns across architectures to characterize how synthetic image mixing influences deep neural network training.

Research Papers

Institutional DEI Discourse in US Higher Education Amid Political Backlash [Working Paper]
Nur Ahmed, **Sabbir Ahmed**

Revisiting Image Mixing Augmentations: Inter- and Intra-Class Dynamics [Working Paper]
Sabbir Ahmed, Dr. A F M Shahab Uddin

Research Service

- **Grant Proposal Formulation:** Co-drafted the technical methodology for **SPADE-AI** (**$215K**, ADB), translating research objectives into a concrete hyperspectral imaging pipeline.
- Reviewer at **IEEE COMPAS**

Technical Skills

- **Programming Languages:** Python, R, C/C++, Java
- **ML Frameworks:** PyTorch, OpenCV, YOLO, Hugging Face, Transformers, NLTK
- **Database:** Firebase, MySQL
- **Tools:** Git, Linux/Bash, Docker, Weights & Biases, LaTeX

Work Experience

Teaching Assistant, Department of CSE *Oct 2024 – Present*
Jashore University of Science and Technology *Jashore, Bangladesh*

- Led labs for **200+ students** across **2** academic years in Machine Learning, Data Structures, and IoT, developing course materials, assignments, and assessments.

Network Operations Intern *Feb 2025 – Apr 2025*
Bangladesh Research and Education Network (BdREN) *Dhaka, Bangladesh*

- Monitored network performance across **195 institutions** using Wireshark and SolarWinds.
- Simulated routing and network topologies in Packet Tracer and supported data-center operations.

Erasmus+ MELBU Summer School *Aug 2022*
University of Leipzig, Germany *Leipzig, Germany*

- Completed a **14-day** summer school.

Projects

Bengali License Plate Recognition System *Jan 2024 – May 2024*
Tech Stack: *YOLOv8, PyTorch, Django, OpenCV* *Jashore, Bangladesh*
JUST CSE; ICT Cell JUST

- Achieved **96.2% accuracy** and curated an annotated license-plate dataset in YOLO/COCO formats.
- Deployed on **Raspberry Pi 4** with LiDAR-triggered capture for efficient edge inference.

NASA Space Apps Challenge 2023 *Aug 2023*
Tech Stack: *HTML, CSS, three.js, Python* *Dhaka, Bangladesh*

- Led *Team_Entergalactic* in developing an interactive **3D Mars visualization** application.
- Integrated **9 NASA APIs** covering rover, helicopter, and Mars weather datasets.

Multi-Institutional Aquaculture Research Project *2023 – 2024*
JUST CSE & FMB; Khulna University, FMRT; Ministry of Fisheries and Livestock *Khulna, Bangladesh*

- Engineered a **36-sensor ESP32 array** monitoring six water parameters for saltwater-shrimp research.
- Integrated real-time dashboards and NVR video systems for remote aquaculture monitoring.

Awards and Scholarships

- Scholarship from the Asian Development Bank (ADB) for graduate studies, 2025–2026.
- Finalist, National Innovation Challenge by a2i and UNDP; presented solutions to ministerial panel, 2022.

Standardized Test Score

- **IELTS:** 7.0 (Listening: 7.5, Reading: 7.5, Speaking: 6.5, Writing: 6)